# Jeevasurendar Muthukrishnan

## Senior Backend Engineer | Node.js, AWS Serverless, Distributed Systems

HSR Layout, Bengaluru 560102, India | +91 98657 90261 | jeevasurendar02@gmail.com|
LinkedIn | GitHub | Portfolio

---

## Profile

Senior backend engineer with 5 years of experience owning cloud-native Node.js platforms on AWS from architecture through production. Designed a streaming-grade IPTV microservices platform and an enterprise integration backend, covering serverless APIs, event-driven messaging, caching, security and CDN delivery. Comfortable across the stack with Vue.js and React.js, and works AI-first with Anthropic Claude and OpenAI to ship faster.

---

## Career Highlights

- **Platform architecture:** Designed a cloud-native IPTV microservices platform in Node.js where every service deploys and scales independently.
- **Serverless on AWS:** Built production REST APIs on API Gateway and Lambda, with SQS and WebSockets for asynchronous and real-time communication.
- **Resilience by design:** Engineered retry, failure-handling and fault-tolerant processing into event-driven workflows.
- **Beyond application code:** Ran production CDN infrastructure, including VPN, Linux Netplan networking and CDN-to-origin connectivity.

---

## Technical Skills

- **Languages and frameworks:** Node.js, TypeScript, JavaScript (ES6+), Express.js, Vue.js, React.js
- **Architecture:** Microservices, event-driven design, REST API design, serverless
- **AWS:** Lambda, API Gateway, SQS, EC2, EKS, S3, RDS, IAM, VPC, Route 53, CloudFront, CloudWatch
- **Data and messaging:** PostgreSQL, MongoDB, Redis, Kafka, WebSockets, Nginx caching
- **Delivery and operations:** Docker, CI/CD pipelines, PM2, Linux, Netplan, VPN
- **Security:** JWT, authentication and authorization, IAM least privilege
- **AI-assisted engineering:** Anthropic Claude, OpenAI, prompt engineering, test generation, refactoring

---

## Professional Experience

**Senior Software Engineer** | Infynect Labs Private Limited, Bengaluru | 2021 – 2026

Own backend architecture, delivery and production operations for the company's Node.js platforms on AWS.

**Cloud IPTV Microservices Platform**

**Core Skills:** Node.js, Microservices, AWS Lambda, API Gateway, SQS, WebSockets, Redis, Nginx, CDN, CI/CD, CloudWatch, Linux, VPN

- Architected the platform as independently deployable Node.js microservices, designed for scalability and fault tolerance.
- Delivered the API layer on API Gateway and Lambda, removing always-on servers from the request path.
- Decoupled services with SQS queues and WebSocket channels, with retries and failure handling so that one failing service does not stall the rest.
- Cut repeat load on origin systems with a two-tier cache: Redis for application data and Nginx for HTTP responses.
- Operated the production CDN, configuring VPN, Linux Netplan networking and point-to-point links between CDN and origin servers.
- Automated builds and deployments through CI/CD, and added CloudWatch monitoring for operational visibility.

**Enterprise Backend and Integration Platform**

**Core Skills:** Node.js, Express.js, REST API's, Kafka, PostgreSQL, MongoDB, Event-Driven Architecture, JWT Authentication, Role-Based Authorization, Query Optimization

- Built Node.js and Express.js microservices that connect internal and third-party systems through REST APIs.
- Introduced Kafka for event-driven communication between services, replacing direct service-to-service dependencies.
- Designed PostgreSQL persistence and optimized slow queries to improve API performance.
- Secured every API with JWT-based authentication and role-based authorization.

**Engineering ownership across both platforms**

**Core skills:** AWS EC2, S3, RDS, IAM, VPC, Security Groups, Docker, PM2, CI/CD, Automated Testing**,** CloudWatch, Performance Optimization, Code Reviews

- Locked down AWS environments with IAM, VPC and security groups on least-privilege principles.
- Deployed cloud applications on AWS using EC2, Lambda, S3, RDS and API Gateway.
- Designed AWS architectures for availability, resiliency, performance and cost optimization.
- Containerized backend services with Docker and ran them under PM2 across development and production.
- Implemented CI/CD pipelines with automated testing for builds and deployments.
- Analysed and optimized the performance of backend APIs and database operations.

---

## Education

**Bachelor's degree in Computer Science**
K.S.Rangasamy College of Arts & Science, Periyar University | 2018 – 2021